

PROJECT	Warehouse 60m × 80m	TYPE	Industrial Warehouse / Factory
CLIENT	[Client Name]	LOCATION	Cambodia
CONTRACT PKG	Structural & Civil Works (excl. Steel & Roof)	BUILDING AREA	4,800 m²
ORIGINAL BOQ	\$312,522.00 USD (Rev A — Cost Estimate)	CURRENCY	USD (1 USD = 4,100 KHR, June 2026)

1. EXECUTIVE SUMMARY

This report presents the Senior QS & Commercial Manager's tender review of BOQ Ref MC-BOQ-WH6080-001 (Rev A) for the Warehouse 60×80m project. The original BOQ totals **\$312,522.00 USD** based on cost-estimate rates. Following a comprehensive review of quantities, market rates, risk exposure, and missing scope, the following adjustments have been applied:

- 10% Additional Rate Increase** applied across all applicable items to cover material price escalation, logistics, and contingency.
- 35% Tender Markup** applied to achieve target gross margin of 25–30% after overhead absorption.
- Missing scope items** identified and priced (approx. \$18,500 additional).
- 8 underpriced items** flagged and corrected to market rates.

ORIGINAL BOQ (REV A)

\$312,522

Cost Estimate

AFTER 10% RATE INCREASE

\$343,774

+\$31,252 (10.0%)

+ MISSING SCOPE ITEMS

\$362,274

+\$18,500 added

FINAL TENDER VALUE (35%)

\$489,070

Recommended Submission

2. BOQ REVIEW TABLE — RATE ANALYSIS

ITEM	DESCRIPTION	UNIT	QTY	ORIGINAL RATE	REVISED RATE	+	REVISED AMT	REMARKS
1 — SITE PREPARATION								
1.1	Set-out and surveying	LS	1	1,500	1,800	+20%	1,800	Underpriced; include GPS & benchmark
2 — PILING WORKS								
2.1	Pile supply 300×300, L=6m	m	648	16.00	18.00	+13%	11,664	Market \$17–19/m; price escalation
2.2	Pile drive/install	m	648	10.50	12.00	+14%	7,776	Mobilization cost underestimated
2.3	Pile head cut (108 nos)	nos	108	25.00	28.00	+12%	3,024	Labor-intensive; adjust +10%
2.4	Pile load test (2 nos)	nos	2	800	1,000	+25%	2,000	Subcontractor rate \$900–1,200
3 — EXCAVATION & EARTHWORKS								
3.1	Machine excavation	m³	284.65	8.50	9.50	+12%	2,704	Market rate \$9–11; fuel cost up
3.2	Disposal surplus spoil	m³	150	6.00	7.00	+17%	1,050	Trucking distance variable; risk
3.3	Backfill & compact	m³	151.79	9.00	10.00	+11%	1,518	Material cost increase
3.4	Laterite sub-base (Owner)	m³	1,440	Owner	Owner	—	0	By owner — confirm in contract
3.5	Sand bedding 100mm	m³	480	7.50	8.50	+13%	4,080	Sand price trending up Q3 2026

ITEM	DESCRIPTION	UNIT	QTY	ORIGINAL RATE	REVISED RATE	+	REVISED AMT	REMARKS
4 — LEAN CONCRETE (C10)								
4.1	Lean concrete — pile caps/beams	m³	25.15	90.00	100.00	+11%	2,515	Acceptable; +10% applied
4.2	Lean concrete — sidewalk	m³	14	90.00	100.00	+11%	1,400	Consistent with 4.1
5 — RC PILES (IN PILE RATE)								
5.1	Included in Item 2	LS	1	0	0	—	0	Correct — no duplication
6 — RC PILE CAPS (C30)								
6.1	EC Pile Caps (18 nos)	m³	12.96	165	185	+12%	2,398	C30 market \$175–195; cement up
6.2	C1 Pile Caps (18 nos)	m³	15.55	165	185	+12%	2,877	Consistent with 6.1
6.3	C2 Pile Caps (9 nos)	m³	25.20	165	185	+12%	4,662	Larger caps; formwork complexity
7 — RC TIE BEAMS (C25)								
7.1	Perimeter Tie Beams	m³	42	155	172	+11%	7,224	+10% applied; market acceptable
7.2	Interior Tie Beams Row E	m³	12	155	172	+11%	2,064	Consistent with 7.1
8 — RC PEDESTALS (C25)								
8.1	Pedestals 400×400×500mm	m³	3.60	155	172	+11%	619	Small volume; higher unit cost justified



ITEM	DESCRIPTION	UNIT	QTY	ORIGINAL RATE	REVISED RATE	+	REVISED AMT	REMARKS
9 — GROUND FLOOR SLAB (C25, 150MM)								
9.1	RC C25 slab 150mm concrete	m³	720	150	168	+12%	120,960	Largest cost item; cement & aggregate up 8–12% YoY
9.2	BRC Mesh A10 slab	ton	17.04	1,200	1,350	+13%	23,004	Steel mesh \$1,300–1,400/ton current
9.3	PE membrane 0.2mm	m²	5,040	0.80	0.90	+13%	4,536	Minor; +10% applied
9.4	Saw-cut joints 6×6m	m	2,080	0.80	1.00	+25%	2,080	Underpriced; market \$0.90–1.20/m
9.5	Surface hardener / power float	m²	4,800	2.50	3.00	+20%	14,400	Specialist subcontractor; \$2.80–3.50
10 — REINFORCEMENT STEEL (SD40)								
10.1	Pile rebar SD390	ton	5.89	950	1,080	+14%	6,361	Rebar SD390 market \$1,050–1,150/ton
10.2	Pile cap rebar	ton	6.45	950	1,080	+14%	6,966	Consistent with 10.1
10.3	Tie beam rebar	ton	7.56	950	1,080	+14%	8,165	Consistent with 10.1
10.4	Pedestal rebar	ton	0.47	950	1,080	+14%	508	Small qty — same rate
10.5	Binding wire	kg	305	1.50	1.70	+13%	519	Minor; +10%
11 — FORMWORK								
11.1	FW — Pile caps EC & C1	m²	103.68	17	19	+12%	1,970	Market \$18–22; +10%
11.2	FW — Pile caps C2	m²	50.40	17	19	+12%	958	Consistent
11.3	FW — Tie beams	m²	360	17	19	+12%	6,840	Consistent
11.4	FW — Pedestals	m²	36	17	19	+12%	684	Consistent
12 — PERIMETER SIDEWALK								
12.1	Sidewalk concrete	m³	33.60	120	135	+13%	4,536	Market \$125–140; +10%
12.2	BRC Mesh A7	ton	0.62	1,200	1,350	+13%	837	Consistent with 9.2
12.3	Sidewalk formwork	m	280	6.00	7.00	+17%	1,960	Edge trim material up
12.4	Surface finish broomed	m²	280	2.00	2.20	+10%	616	Standard +10%
13 — MASONRY WORKS								
13.1	200mm block wall	m²	943	27	30	+11%	28,290	Block + labor \$28–32; +10%
13.2	Mortar & accessories	LS	1	800	1,000	+25%	1,000	Underpriced for 943 m² wall
14 — PLASTERING & FINISHES								
14.1	Internal plaster 20mm	m²	943	7.00	8.00	+14%	7,544	Labor-intensive; +10%

ITEM	DESCRIPTION	UNIT	QTY	ORIGINAL RATE	REVISED RATE	+	REVISED AMT	REMARKS
14.2	External plaster 20mm	m²	943	8.50	9.50	+12%	8,959	Scaffold required; +10%
14.3	Internal paint	m²	943	4.50	5.00	+11%	4,715	Standard +10%
14.4	External paint	m²	943	5.50	6.20	+13%	5,847	Weather-grade paint premium
15 — OPENINGS								
15.1	Roller shutters 4.0×4.5m	nos	8	2,100	2,400	+14%	19,200	Motor + install \$2,200–2,600 ea.
15.2	Steel personnel door	nos	4	40	120	+200%	480	CRITICAL: \$40 unrealistic; painted steel door \$100–150
15.3	Aluminium glass window	m²	129.60	80	95	+19%	12,312	Market \$85–110/m² installed

Revised Direct Cost (after 10%+ rate adjustment)

\$343,774 USD

3. RISK ASSESSMENT

#	RISK DESCRIPTION	CATEGORY	LEVEL	MITIGATION
R1	Ground slab concrete (Item 9.1) is 35% of total — cement/aggregate price volatility	Commercial	HIGH	Include price escalation clause in contract; fix supply price with batching plant
R2	Rebar SD390 (Items 10.1–10.4) — steel market fluctuation risk	Procurement	HIGH	Lock supply contract within 30 days; hedge via forward purchase
R3	Laterite sub-base (Item 3.4) by Owner — delay risk if owner procurement is late	Programme	MED	Contractual provision for EOT if owner-supplied material is delayed
R4	Piling — unforeseen ground conditions; pile length may exceed 6m	Technical	HIGH	Provisional sum for extra pile length; confirm with soil investigation
R5	Steel personnel door priced at \$40 — loss-making item	Commercial	HIGH	CORRECTED to \$120/door in revised BOQ
R6	Roller shutter motor-operated — import lead time 6–8 weeks	Procurement	MED	Order upon contract award; confirm specifications early
R7	Rainy season (May–Oct) — earthworks & concrete productivity impact	Construction	MED	Programme earthworks for dry season; rain protection for concrete
R8	No soil investigation report available — pile design assumptions	Technical	HIGH	Conditional tender; exclude redesign liability
R9	Scaffold for external wall works (9m eave) not priced separately	Technical	MED	Included in revised plaster/paint rates; add provisional item
R10	BRC Mesh import dependency — supply chain disruption	Procurement	LOW	Alternative local suppliers identified; 2-week buffer stock

4. MISSING SCOPE ITEMS IDENTIFIED

ITEM	DESCRIPTION	UNIT	QTY	RATE	AMOUNT	JUSTIFICATION
M1	Dewatering during excavation (rainy season)	LS	1	3,500	3,500	Not priced; essential if excavation during May–Oct
M2	Scaffold & temporary works for wall finishes (9m ht)	LS	1	4,500	4,500	External plaster/paint at 9m eave requires scaffolding
M3	Concrete curing compound / water curing for slab	m²	4,800	0.50	2,400	4,800 m² slab requires proper curing — not in BOQ
M4	Waterproof membrane at pile cap/pedestal junction	m²	200	3.50	700	Good practice for foundation waterproofing
M5	Expansion joints at slab perimeter (compressible filler)	m	320	4.00	1,280	Perimeter isolation joints missing from BOQ
M6	Lintel beams over door/window openings (RC)	nos	48	45.00	2,160	48 openings (8 shutters + 4 doors + 36 windows) need lintels
M7	Column/wall connection dowels (starter bars)	nos	45	15.00	675	45 pedestals need dowels into masonry wall
M8	Site mobilization / demobilization	LS	1	2,500	2,500	Equipment transport, temp facilities, site setup
M9	Quality testing (concrete cube tests, slump)	LS	1	800	800	Minimum 20 cube tests for 750+ m³ concrete

Total Missing Scope Items

\$18,515 USD

5. UNDERPRICED ITEMS FLAGGED

ITEM	DESCRIPTION	ORIGINAL RATE	MARKET RATE	REVISED RATE	ISSUE
15.2	Steel personnel door 1.0×2.1m	\$40	\$100–150	\$120	CRITICAL — 200% below market. Possible typo in original BOQ.
9.4	Saw-cut joints	\$0.80/m	\$0.90–1.20	\$1.00	Below specialist subcontractor rates
2.4	Pile load test	\$800/test	\$900–1,200	\$1,000	Dynamic test subcontractor rates higher
13.2	Mortar mix & accessories	\$800 LS	\$1,000–1,200	\$1,000	Insufficient for 943 m² blockwork
9.5	Surface hardener / power float	\$2.50/m²	\$2.80–3.50	\$3.00	Specialist trade — subcontractor premium
1.1	Set-out and surveying	\$1,500 LS	\$1,800–2,500	\$1,800	GPS equipment, licensed surveyor, benchmarks
10.1–4	Rebar SD390 (all items)	\$950/ton	\$1,050–1,150	\$1,080	Steel price escalation Q2–Q3 2026
15.3	Aluminium glass window	\$80/m²	\$85–110	\$95	6mm tempered glass + powder-coat frame

6. FINAL TENDER PRICING SUMMARY

COMPONENT	AMOUNT (USD)	% OF TENDER	NOTES
A. Original BOQ (Rev A — Cost Estimate)	312,522	64.0%	As submitted by design team
B. Rate Adjustment (+10% & market correction)	31,252	6.4%	Material escalation, underpriced items corrected
C. Missing Scope Items (M1–M9)	18,515	3.8%	Dewatering, scaffold, curing, lintels, mob/demob
D. Revised Direct Cost (A + B + C)	362,289	74.1%	Base cost for markup calculation
E. Overhead & Profit (OH&P) @ 15%	54,343	11.1%	Head office OH, insurance, bonds, profit
F. Contingency @ 5%	18,114	3.7%	Unforeseen ground conditions, weather delays
G. Additional Markup to 35% target	54,324	11.1%	Commercial margin to achieve tender target

Revised Direct Cost (D)	\$362,289 USD
OH&P + Contingency + Markup (E + F + G)	\$126,781 USD
RECOMMENDED FINAL TENDER VALUE	\$489,070 USD

Commercial Manager's Recommendation

Recommended Tender Submission Value: \$489,070 USD (excl. steel structure & roof)

This represents a 35% markup on the revised direct cost, yielding an estimated gross margin of 26–28% after overhead absorption. The tender remains competitive within the Cambodia industrial warehouse market range of \$95–115/m² for civil & structural works (this tender: \$101.9/m²).

Key conditions for submission:

- 1. Include price escalation clause for concrete and steel (items exceeding 60 days from award).
- 2. Confirm soil investigation report — tender is conditional on 6m pile length assumption.
- 3. Laterite sub-base by Owner — include EOT provision for late supply.
- 4. Lock rebar supply price within 14 days of contract award.
- 5. Provisional sum of \$5,000 recommended for extra pile length (if ground conditions differ).
- 6. Validity period: 90 days from submission date.

RECOMMENDED FOR SUBMISSION

Prepared By Senior Quantity Surveyor	Reviewed By Commercial Manager	Approved By Managing Director
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